

COMP30040: Third Year Project

Guidance on Report Writing

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1 Introduction

This guide gives you some tips on writing your third year project report and it complements the information that is available on Blackboard.

A lot of this guide concerns $\text{\LaTeX}$ but there are some comments on writing your report in general.

★ The advice given here and on Blackboard is not the be-all and end-all. Your supervisor may advise you to do things differently. Seeing as your supervisor marks your work, it is probably best to take their advice.

2 Microsoft Word or $\text{\LaTeX}$?

You can write your report using whichever tool you like. But which should you choose?

You might be familiar with Word, but you should also be familiar with $\text{\LaTeX}$ through work you have done on your degree. If you want to publish papers in the future, maybe it is better to get more familiar with $\text{\LaTeX}$. If you intend to go into industry where using Microsoft Office (or a clone) might be the norm, then perhaps it will be of benefit to delve deeper into Word.

A lot of the things that you will need for your report are easy to do with $\text{\LaTeX}$, such as a table of contents; figures and tables; mathematics; cross-referencing; citations; and so on. I believe that things like this are a bit more difficult in Word because they are not the sort of thing that the typical Word-user needs. Therefore, if you choose Word, you will need to work out how to do these things. If you want to know how to do something in $\text{\LaTeX}$, it is most likely that your supervisor will be able to help you.

I love $\text{\LaTeX}$ and would recommend it for your report. I think that a lot of the work in making an attractive, professional-looking document is done for you. But, as already stated, the choice is entirely yours.

3 Getting Started

There are videos on Blackboard that gives you advice on how to get started. A common method is to set up your document with chapter and section titles. For the chapter titles, you might choose Introduction, Background, Design, Testing, Conclusions, for example. You then think about all of the things that you need to talk about in these chapters. In your Background chapter, if you are doing a machine learning project, you might need to talk about deep learning, amongst other things. So, you might have a Deep Learning section, and within that have sub-sections for things like basic theory, datasets, architectures, and so on.

Your writing tool will then produce a table of contents that you can look at and think about other sections you might need, or if the order needs changing. Remember that you can add and move sections at any time; this is just to get something written down.

In $\text{\LaTeX}$, you add chapters and sections as shown below.

```

\chapter{Background} \label{cha:background}

% Using \label{} allows you to reference the chapter elsewhere with \ref{cha:background}

\section{Deep Learning}

\subsection{Basic Theory}

\subsection{Datasets}

\subsection{Neural Network Architectures}

```

Each chapter is usually placed in a file of its own and they are all *included* (`\include{filename.tex}`) in a main file called, for example, `report.tex`.

On Blackboard, a couple of L^AT_EX **templates** are provided and these set up the general layout and formatting of the document. You can then add your chapter files and section titles and get writing.

It might help you to write notes in each section reminding yourself of what you want to say. You can keep adding notes as you think of things. Often, when you are properly writing a section, you will be reminded of something that you need to write in another section. You can just go to the relevant section and write a note and then go back to the writing you were in the middle of.

With all of these notes dotted around, it is easy to forget about them and leave them in your final version as they are. I like to write ‘TODO’ with my notes so that I can later search for all of the TODOs and sort them out.

4 Word Counts

You might start off writing your Background chapter and before you know it, you have written 5000 words. That is about half of the required number of words for the whole report. Keep an eye on the word count as you write. If you write too much, you might end up whittling it down later — and that’s a pain.

You could set out some rough proportions for the different chapters in your report and, using a target number of words, for example 12000, make an estimated word count for each chapter. An example is shown here:

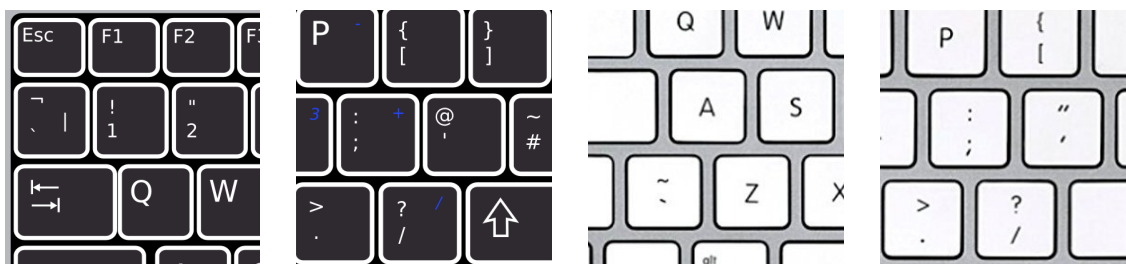
Chapter	Proportion	Number
Introduction	10%	1200
Background	30%	3600
Design	25%	3000
Results	25%	3000
Conclusion	10%	1200

If you write your L^AT_EX document on the Overleaf website, you can get a word count for a chapter by selecting it in the menu item for word counts.

If you use the *uthesis* template that is available on Blackboard, the word count is written on the last contents page. However, this number is not updated automatically; the number is read from a file named `word.count`. You need to manually obtain a word count and then write the number in that file.

5 Quotation Marks

In your report, quotation marks should preferably look like a little 66 and 99 (or 6 and 9 for single quotes). For example, ‘some quote’ and “another quote”.



(a) PC backtick next to the '1' key (b) PC quote on the '@' key (c) Mac backtick next to the 'Z' key (d) Mac quote on the double-quote key

Figure 1: Backtick and quote keys.

In $\text{\LaTeX}$, you don't use the double-quote key, you use a single back-quote (backtick) to open quotes and a single normal quote to close quotes. You use these either singly or in pairs:

Here is something in single quotes: ``some quote'`.
 And here is something in double quotes: ```another quote''`.

Figure 1 shows where you might find your backtick and single-quote keys on your keyboard.

6 Referencing

You can give a label to lots of things in your document using `\label{}` — chapters, sections, figures, tables, equations, and so on. For example:

```
\section{Deep Learning} \label{sec:DeepLearning}
```

It is common to put a prefix on your labels with a colon, for example, 'fig:' for figure, 'sec:' for section, 'eq:' for equation.

You can reference labelled objects elsewhere in your document using `\ref{}`. For example:

```
See Section~\ref{sec:DeepLearning}.
```

In the above example, a tilde (`~`) is used. The tilde signifies a non-breaking space. This prevents a line-break in the middle of your reference. That is, it will stop 'Figure' appearing at the end of a line, and '2.1' (for example) appearing at the start of the next line.

When referring to sections, figures and so on, use a capital letter. You are referring to the name of a particular section or figure, so write 'see Figure 2.1' rather than 'see figure 2.1'.

7 Figure and Table Captions

For all of your captions, try to write a title and a description. For example:

```
\caption{GPS satellite constellation. The American GPS systems currently consists of 31 satellites orbiting the Earth.}
```

Imagine that somebody is scanning your report to see if it will be of use to them. They might quickly look through the document at the figures (a picture paints a thousand words). If they see a figure that might indicate that your work is relevant, they wouldn't want to waste time looking through your text to find where you referenced the figure. They would want to read the caption. So, you need to describe in the caption what

you are showing with the figure.

Also, if the figure contains an image that you got from somewhere, cite the source as in the following examples.

```
\caption{GPS satellite constellation. The American GPS systems currently consists of 31 satellites
    orbiting the Earth. Source [Online] -- available at: \url{https://www.gps.gov/}.}

\caption{GPS satellite constellation. The American GPS systems currently consists of 31 satellites
    orbiting the Earth. Source: [12].}
```

The muthesis template adds a List of Figures and List of Tables near the top of the document. If you write your captions as shown above, the whole of that text will appear in the lists. It is usually better to only show the title in the lists. This can be done by using the *short text* option on the `\caption{}` command as shown below.

```
\caption[short text]{Long text}
```

The short text appears in the List of Figures, the long text appears on the figure. Using the above example, you would write:

```
\caption[GPS satellite constellation.]{GPS satellite constellation. The American GPS systems
    currently consists of 31 satellites orbiting the Earth. Source [Online] -- available at:
    \url{https://www.gps.gov/}.}
```

8 Sub-figures

You often want to put more than one diagram or image in a figure. This is done in Figure 1. You could just do this by putting multiple diagrams or images in the figure yourself, but you have to do extra work to give them their own captions and to be able to reference them like Figure 1a and Figure 1b.

The easiest way to handle multiple figures (sub-figures) is to use the `subcaption` package. You need to add the following to your document preamble (in `report.tex` if you are using the muthesis template).

```
\usepackage{subcaption} % for sub-figures
```

Now, within your figure environments, you can use `subfigure` as shown below.

```
\begin{figure}
\centering

\begin{subfigure}{0.22\linewidth}
\centering
\includegraphics[width=0.9\textwidth]{figures/pc_backtick}
\caption{PC backtick next to the `1' key}
\label{fig:pc_back}
\end{subfigure}
%
\begin{subfigure}{0.22\linewidth}
\centering
\includegraphics[width=0.9\textwidth]{figures/pc_quote}
\caption{PC quote on the '@' key}
\label{fig:pc_quote}
\end{subfigure}

\caption{Backtick and quote keys.}
\label{fig:keys}
\end{figure}
```

You don't need to add a caption to each subfigure; you can just write `\caption{}` and the subfigure will be labelled with the '(a)' and '(b)' alone.

9 Numbers

If you mention a quantity in your text, don't write something like 'There are 2 ways of doing this.' Instead, spell the word out: 'There are two ways of doing this.'

You obviously wouldn't do this when the numbers are large, for instance: 'There are 7000 languages spoken around the world.' Nor would you do it when writing measurements such as 'a positional accuracy of 2 m' unless you also spell out the units: 'a positional accuracy of two metres.'

Incidentally, when writing numbers with units, you don't normally write the units right next to the number as in '2m'. But if you put a space between them, there is a chance that a line split occurs between them. It is therefore better to use a non-breaking space (`2~m` produces 2 m) or (even better) a shorter non-breaking space (`2\,m` produces 2 m).

10 Size of Figures

Don't make images and diagram too big or too small. If there is text in the image that you want the reader to read, aim for it to appear about the same size as the main text or slightly smaller.

Rather than using the `scale=0.5` option which you would have to change for each image dependent on its resolution, you can specify the width of the image as a proportion of the width of text on the line:

```
\includegraphics[width=0.5\textwidth]{some_image}
```

You might also want to centre your images on the page (if not using a floating *figure* environment):

```
\begin{center}  
  \includegraphics[width=0.5\textwidth]{some_image}  
\end{center}
```

11 Contractions

Your project report should be a formal document, so don't use contractions. As an example, write 'they are' rather than 'they're' and 'does not' rather than 'doesn't'.

Also, be wary of using forms of shorthand; write 'Results and Evaluation' rather than 'Results & Evaluation'.

If you are unsure of how particular things are normally done in academic writing, just look in a computer science book or a quality academic paper.

12 Hyperref Package

Nowadays, your $\text{\LaTeX}$ document is more likely to be read as a pdf rather than being printed onto paper. To make your document easy to navigate, it is a good idea to use the **hyperref** package. Add this in your document preamble:

```
\usepackage{hyperref} % for clickable links
```

Whenever you use `\ref{}`, `\cite{}` or `\url{}`, this package creates a hyperlink to whatever is being referenced, for example a figure or a bibliography entry. Depending on the pdf viewer that is being used to read the document, hovering on the link may display a preview of the destination so that the reader doesn't have to actually click on the link.

A drawback of hyperref is that, by default, hyperlinks are drawn with (in my opinion) an ugly, tight rectangle around the text:

. This is done in this document in Figure [1](#)
 the figure yourself, but you have to do extra
 hem like Figure [1a](#) and Figure [1b](#)

If you don't like how the links are formatted, you can change them. Using the following settings:

```
\usepackage{hyperref} % for clickable links
\hypersetup{
  colorlinks,
  citecolor=blue,
  filecolor=blue,
  linkcolor=blue,
  urlcolor=blue
}
```

makes links look like this:

. This is done in this document in Figure [1](#).
 the figure yourself, but you have to do extra
 hem like Figure [1a](#) and Figure [1b](#).

13 BibTeX

You might notice that author names, written in your bib file in full, get abbreviated in your document. For example,

```
author={Abbas, Ali and Jafar, Assef and Dahrouj, Zouhair},
```

ends up as 'A. Abbas, A. Jafar, and Z. Dahrouj'. This is fine. But what if the author is written as:

```
author={World Health Organization},
```

You wouldn't want this abbreviated to 'W. H. Organization'. To prevent this, surround the text that you don't want to be abbreviated with an extra set of braces:

```
author={{World Health Organization}},
```

Something similar is done in citation titles where it tries to get rid of capitalised words. For instance a title like `title={GPS/IMU data fusion ...}` appears as 'Gps/imu data fusion ...'. To avoid this, surround the abbreviations with braces: `title={{GPS/IMU} data fusion ...}`.

14 Multiplication Symbol

When people type image sizes, for example 800 by 600, they write 800x600, 800X600, or 800*600. It would probably look nicer to use the `\times` symbol within mathmode:

`$800 \times 600$` gives 800×600 . `800$\times$600` gives 800×600 .

15 Tables

You use the *tabular* environment to draw tables. The word-count table in this document was written as:

```
\begin{tabular}{l c c}
\hline
Chapter & Percentage & Number\\
\hline
Introduction & 10 & 1200\\
Background & 30 & 3600\\
Design & 25 & 3000\\
Results & 25 & 3000\\
Conclusion & 10 & 1200\\
\hline
\end{tabular}
```

The ‘l c c’ specifies the alignment for each column in the table (l = left, c = centre). You might also want to specify a fixed-width column with something like ‘p{2cm}’.

If you want your table to have a caption and to be able to reference it, you need to put it in a *table* environment:

```
\begin{table}
\centering
\begin{tabular}{l c c}
\hline
Chapter & Percentage & Number\\
\hline
Introduction & 10 & 1200\\
Background & 30 & 3600\\
Design & 25 & 3000\\
Results & 25 & 3000\\
Conclusion & 10 & 1200\\
\hline
\end{tabular}

\caption{Estimated word counts for each chapter.}
\label{tbl:proportions}
\end{table}
```